

YOURSTORY



1126



CoCo CLI

HACKATHON

GCC Edition



Team Name : Aidhunik India

Team Leader Name : Paras Lehana

Team Members : Paras Lehana (team leader), Abhinav Kaushik

Team Size : 2

Problem Statement : Predictive Maintenance and OEE Command Center

Nirantar (निरंतर)

AI Reliability Command Center on Snowflake, built end to end with CoCo

From a vibration spike to an approved, parts-ready work order in minutes.
All data synthetic (fictional "Indus Group").

1 • PROBLEM BRIEF

Unplanned downtime is the most expensive minute in manufacturing

OT signals live in SCADA/historians, spares and orders in ERP, history in the CMMS, fixes in PDF manuals. Alerts arrive without context, so teams react after the line stops.

\$1.4T

lost yearly by the Fortune Global 500 to unplanned downtime (11% of revenue). Siemens 2024

\$2.3M/h

cost of an idle automotive line. Siemens 2024

88%

of Indian industrial firms have an outage at least monthly, ≈ ₹70 lakh/h. ABB 2023

~60%

typical OEE vs 85% world class

Who feels it

- **Reliability engineer:** finds failing assets too late
- **Maintenance planner:** hours chasing parts, people and a window
- **Plant manager:** owns OEE and delivery dates

Pain today

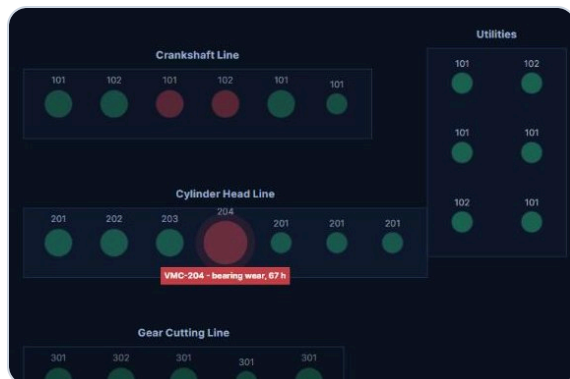
- Alarm floods, sensor faults treated as failures
- No link from vibration to spares, schedule, cost
- Root cause buried in notes, lots and shifts

Domain context

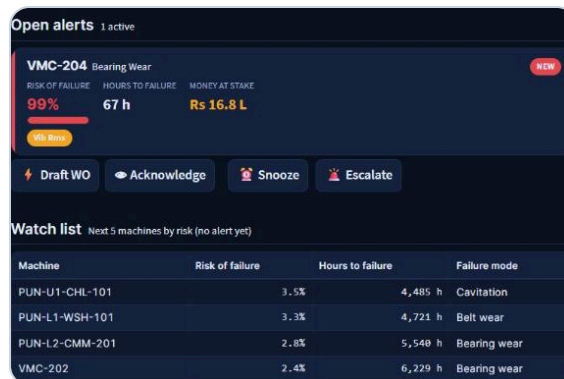
- ISA-95 plant hierarchy, ISO 14224 failure taxonomy, ISO 10816 vibration zones
- SAP PM / Maximo-style work orders
- India's 2,117 GCCs run reliability analytics for global plants

2 · NIRANTAR IN ONE LOOP

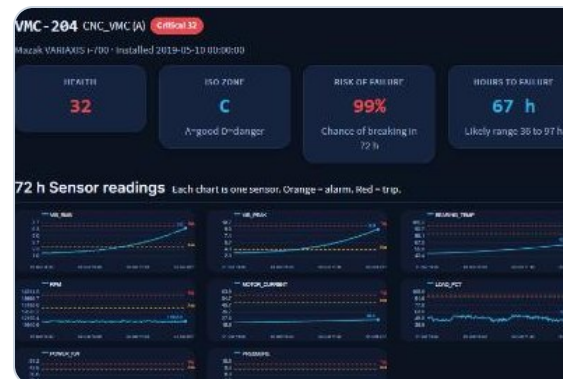
Predict → prioritise → explain → act → approve



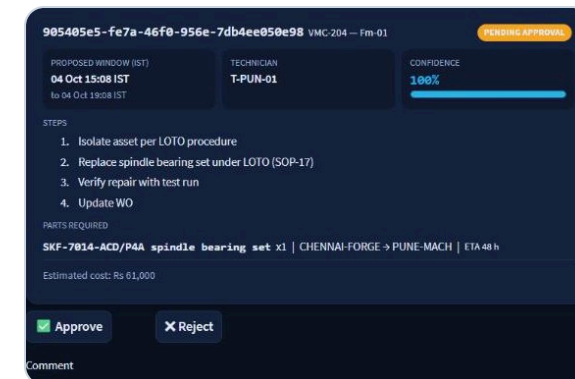
1 **See.** Live twin: VMC-204 turns red (bearing wear, ISO zone C, RUL 67 h).



2 **Prioritise.** Risk 0.99, RUL 67 h and about ₹16.8 lakh at stake rank it #1 (risk × ₹/h × order pressure). Sensor faults are suppressed.



3 **Explain.** Vibration is 4.1× normal and crossed the alarm line; bearing temperature is climbing; all related sensors agree, so it is bearing wear, not a sensor fault. The Cortex Agent cites the SOP and stock.

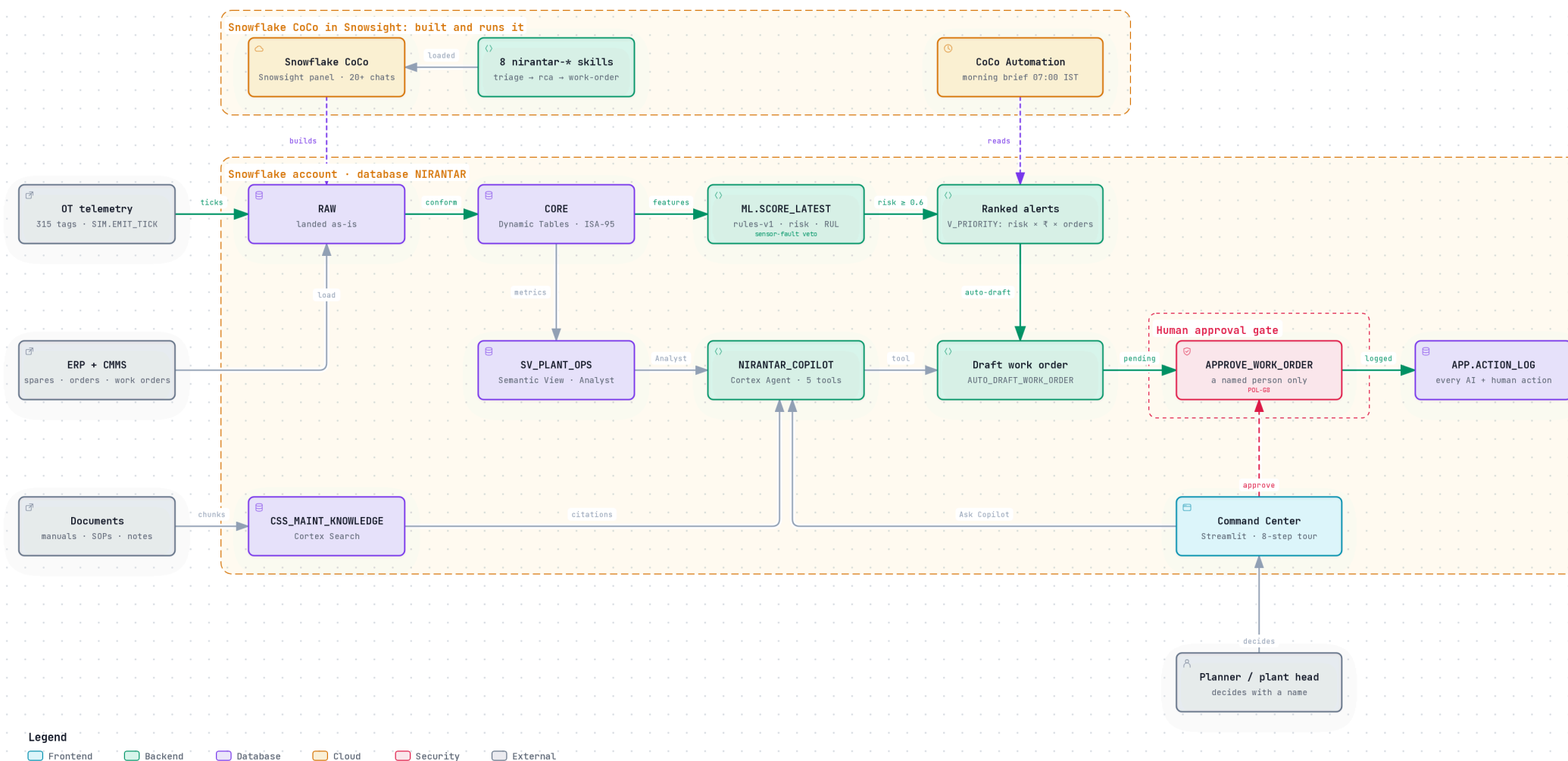


4 **Act.** Draft WO: bearing transferred Chennai → Pune, earliest low-impact window before the predicted failure, certified technician, LOTO steps. A **human approves**; audit logged.

IT + OT converged in one governed Snowflake account: OT telemetry (vibration, temperature, RPM, current, pressure) ⋈ ERP (spares by plant, supplier lots, production & sales orders) ⋈ CMMS (work orders, failure codes, technicians) ⋈ documents (manuals, SOPs, Hinglish notes).

3 · ARCHITECTURE DIAGRAM

Snowflake-native, modular, built by CoCo



4 · BUILT WITH COCO ACROSS THE LIFECYCLE

Every object was planned, built, run and tested by CoCo

Phase	What CoCo did
Planning	Framed the problem, drew the ontology (ER) and workflow, mapped components to skills, profiled the data
Development	Synthetic enterprise with failure physics, Dynamic Tables, scoring, agent tools, semantic view, search, Cortex Agent, Streamlit app
Execution	Serverless tasks + alert, hero scenario run, scheduled CoCo Automation (morning briefing)
Testing	Integrity/physics tests (21 of 22 (the 22nd, agent pass rate, is reported separately)), agent golden set (60 % (9/15)), edge cases
Surfaces	CoCo in Snowsight (Cloud Agent) · CoCo CLI v1.1.104 installs the 8 skills from GitHub · Snowflake CoWork

Evidence: 20+ chats CoCo sessions · journal + screenshots + usage export in the repo.

Done. Here's the summary:

- 1. Written `nirantar/docs/coco-planning.md` (437 lines) with all 5 sections: 10-bullet problem framing (personas, decisions, cost of not knowing), Marmaid erDiagram (ISA-95 + ISO 14224 + ERP + CMMS ontology), Marmaid flowchart (tick → features → health/RUL → alert → WO draft → approval → audit → OEE), 21-row build-plan table mapping every component to its Snowflake feature and CoCo skill, and 8 risks with mitigations and test plans.
- 2. Key architectural decisions captured: DTs at ≥15-min lag (not 1-min, to avoid 100/day burn), serverless tasks for the live scoring path (~3/day), warehouse-runtime Streamlit via SQL CREATE STREAMLIT, agent call via DATA_AGENT_RUN SQL with CoWork fallback.
- 3. Build order is clear: enterprise master data → history → docs → simulator → pipeline DTs → features → sensor-fault detector → health/RUL → alerts → WO automation → semantic view → search → agent → Streamlit → automation → governance → tests → evidence.
- 4. Journal updated: entry #2 in `NIRANTAR.OPS.COCO_JOURNAL` (phase PLANNING, artifact `nirantar/docs/coco-planning.md`).
- 5. No Snowflake objects were created — this is a planning-only deliverable. Next phase: DEVELOPMENT, starting with `/nirantar/synthetic-plant` to generate the Indus Group enterprise.

Planning (CoCo in Snowsight): read our specs, wrote the plan, ontology and build order

8 reusable CoCo skills (headline bonus)

nirantar-synthetic-plant

nirantar-ontology-semantic-view

nirantar-command-center

nirantar-alert-triage

nirantar-rca

nirantar-work-order

nirantar-reliability-brief

nirantar-coco-evidence

Chained workflow (input → processing → output):

`/nirantar-alert-triage PUN-L2` → `/nirantar-rca VMC-204` → `/nirantar-work-order VMC-204`

Installed in the workspace (`.snowflake/cortex/skills`), versioned in `plugin.json`, MIT-licensed on GitHub.

Guardrails

Human approval gate · sensor-fault veto · idempotency + rate limits · confidence floor · CoCo daily credit caps · resource monitor · RBAC · full audit log

Test results (top-2 titles per query):

Query	#1	#2
spindle bearing noise + high temp after greasing	OEM Manual — CNC VMC (MANUAL)	OEM Manual — CNC VMC (MANUAL)
LOTO steps for CNC spindle bearing replacement	SOP-18: Spindle Bearing Replacement (SOP)	SOP-18: Spindle Bearing Replacement
supplier lot hardness deviation	Shree Bearings Lot L-2391 (SUPPLIER_REPORT)	Hydraulic Seal Kit Variance (SUPPLIE
compressor trips in hot weather	CHIT-L1 Night Shift Log (SHIFT_LOG)	EB-2026-04 Compressor Heat Dera
Line 2 machine bear bear band ho rahi hai	PUN-L2 Shift B Hinglish log (SHIFT_LOG)	PUN-L2 Shift C Hinglish log (SHIFT_L

All queries return semantically correct results. The Hinglish query correctly surfaces translated Hinglish shift logs about VMC-204

Development (CoCo in Snowsight): built Cortex Search over 34 documents and tested 5 queries, incl. Hinglish

5 · MEASURED RESULTS (SYNTHETIC DATA, MEASURED IN SNOWFLAKE)

Measured, not claimed

83.7 %

historical failures detected early (backtest)

36 h

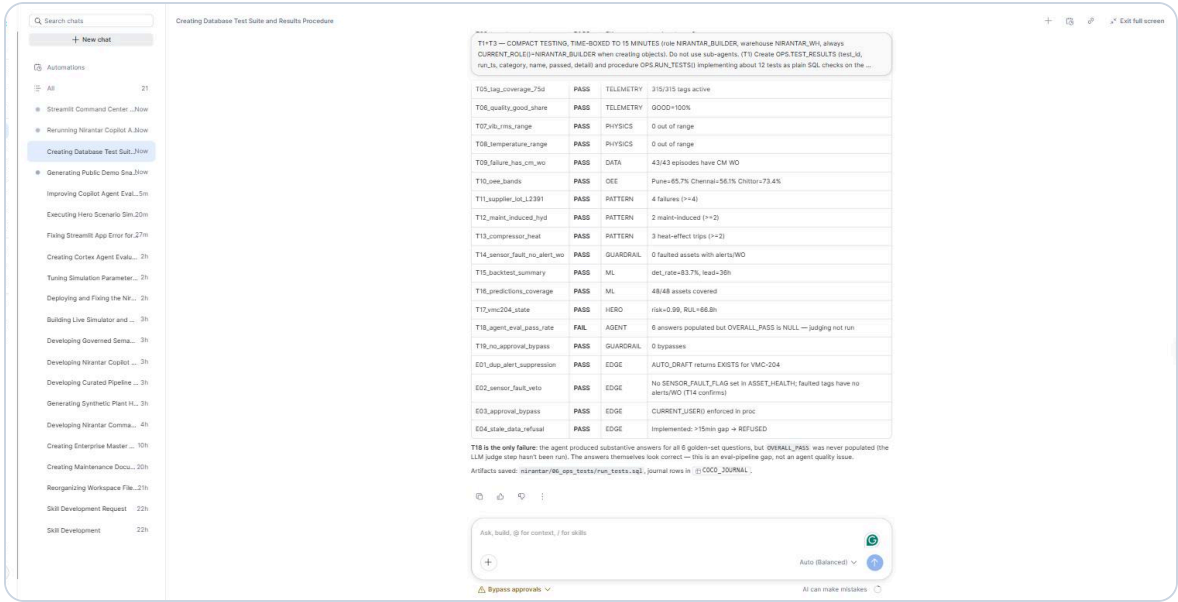
median early-warning lead time

0.28

false alarms per asset per month

60 % (9/15)

agent golden-set pass rate



Metric	Value
Telemetry rows / tags / assets	6.8 M / 315 / 48
Failure episodes in history	43
OEE by plant (last 30 d)	Pune 65.7 % · Chennai 56.1 % · Chittor 73.4 %
Unplanned downtime cost (75 d)	see OPS tables
Sensor-fault false dispatches prevented	veto verified in the flat-line demo
Data tests passed	21 of 22 (the 22nd, agent pass rate, is reported separately)
Alert → approved WO (demo)	< 1 min (6 ticks, auto-drafted)
Live-path run cost	≈ \$3/day serverless; DTs at 15-min lag

6 · IMPACT STATEMENT

Fewer breakdowns, faster decisions, higher OEE

Measurable outcomes

- PdM cuts machine downtime **30–50%** and extends life 20–40% (McKinsey); predictive-led plants see **52.7% less** unplanned downtime (NIST)
- WO preparation: hours of cross-system chasing → **one approval**
- Every downtime minute priced in ₹, plus energy and CO₂ waste of degraded assets

Scalability

- One ontology + semantic view: every surface returns the same OEE
- Industry packs (auto components, cement, FMCG) from one generator skill
- Snowflake scales from 3 plants to 300 with no data movement; serverless live path

Beyond the demo

- Swap synthetic RAW for OPC-UA/historian (Snowpipe Streaming) and SAP PM/Maximo feeds
- Already in the public demo: technician job card (EN / हिन्दी), maintenance calendar, root cause, alarm rules, business case
- Jira / Slack / Drive via MCP: payloads previewed in the demo, live wiring next; multi-agent war room

Pilot hypothesis: one critical line, 6 weeks, success = $\geq 50\%$ of breakdowns flagged ≥ 24 h ahead, ≤ 1 false alarm/asset/month, and every WO approved with parts on hand.

7 · HOW NIRANTAR MEETS THE BRIEF AND THE COCO GUIDELINES

Every point of problem #03, with the proof

The brief / guideline	How Nirantar does it	Where to see it
Converge IT and OT data	OT ticks, ERP, CMMS and 34 documents in one Snowflake account, conformed into an ISA-95 ontology by Dynamic Tables	Architecture slide · demo <i>How it is built</i>
Correlate vibration, temperature, RPM with ERP + maintenance	1-minute ticks → 15-minute features → V_PRIORITY joins risk to ₹/hour, orders due, stock and CMMS history; one IT + OT timeline per machine	Demo: Machine Detail
Predict failures in advance	Health, ISO 10816 zone, 72 h risk, time to failure with a range; back-test: 36 of 43 caught, median 36 h early, 0.28 false alarms / machine-month	Demo: Trust Audit · evidence T/01
Root cause in natural language	Cortex Agent: Analyst on SV_PLANT_OPS (verified queries) + Search over manuals and Hinglish notes + 5 tools; every answer cites sources	Demo: Ask Copilot, Reliability
Command center: triage and action	Alerts ranked by confidence × criticality × ₹ × order deadline; acknowledge / shelve / escalate; technician job card	Demo: Alert Triage, Technician
Automate work orders	Alert → AUTO_DRAFT_WORK_ORDER (spares across plants, window before failure, technician, SOP steps) → APPROVE_WORK_ORDER by a named person	Demo: Work Orders · evidence E/01
Lift OEE	A × P × Q by line and shift, six big losses; the planned repair recovers 46 h and Rs 16.2 L	Demo: OEE, Business case
CoCo: plan, build, run, test	20+ CoCo chats in Snowsight; CoCo-written tests and golden set; CoCo Automation 07:00 IST	docs/evidence (P, D, E, T)
Ingenuity	8 reusable skills (installable with the CoCo CLI from GitHub), 5 custom agent tools, automation, guardrails; MCP payloads for Jira / Slack / Drive previewed (not wired yet)	skills/ · evidence X/02 · demo Work Orders

The same map is a page in the public demo: **For judges: brief coverage** (#/judges). All data synthetic.

8 · ROADMAP AND HOW TO TRY IT

Prototype today, finale-ready next

Next for the finale

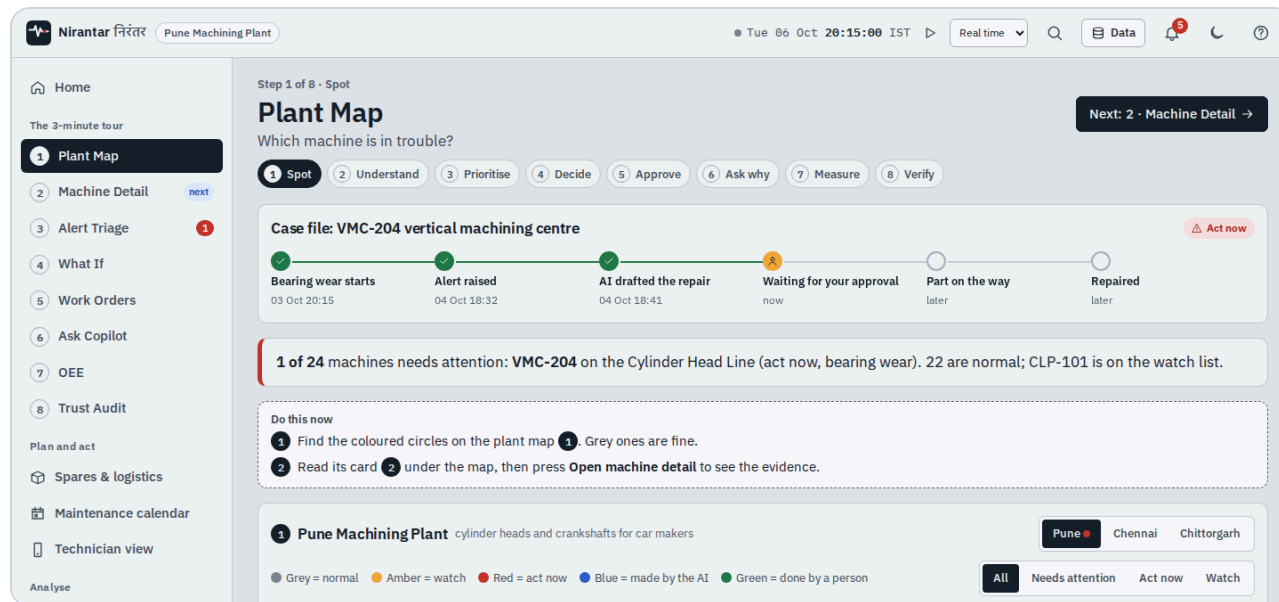
- ML classifier + SHAP explanations blended with physics rules
- Failure Time Machine: replay the 72 h before a breakdown on the twin
- Multi-agent War Room (signal, history, supply analysts → RCA synthesiser)
- Managed MCP server + CoCo CLI/Desktop and CoWork across surfaces
- Public live link; second industry pack; governance policies live

Try it

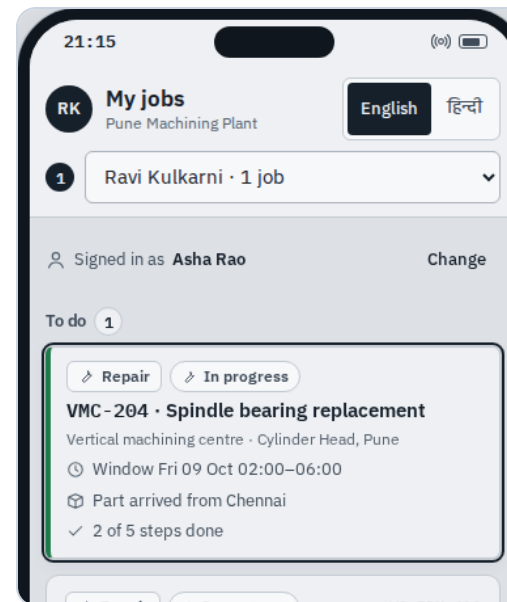
- Video: (YouTube link in the submission form)
- Repo: github.com/paras-lehana/nirantar
- Deployed demo: paras-lehana.github.io/nirantar/app (interactive demo, no login)
- Rebuild: install the skills, then `/nirantar-synthetic-plant` and follow the runbook

APPENDIX · TRY IT WITHOUT A LOGIN

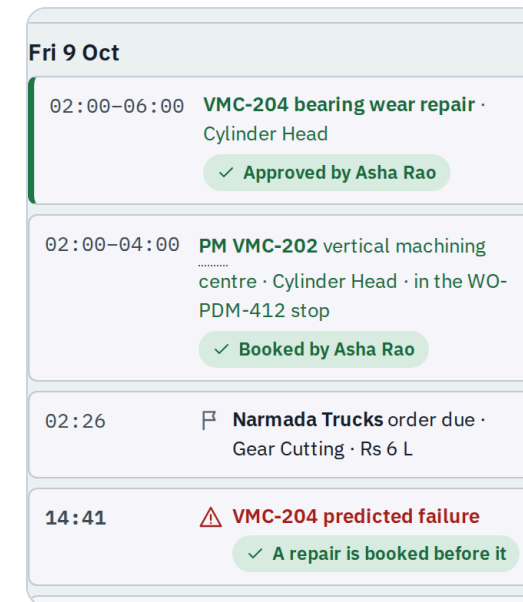
The public interactive demo runs the whole loop in your browser



1 · Plant Map: grey means normal, colour means act. VMC-204 red, CLP-101 amber.



Technician job card: lock-out first, release reading ≤ 2.3 mm/s.



Calendar: overdue jobs bundled into a planned stop.

paras-lehana.github.io/nirantar/app (interactive demo, no login) · 8-step guided tour + 14 tools (calendar, technician view, root cause, peer comparison, alarm rules, business case, spares, energy, shift brief, presenter) · 5 scenarios with Load sample data · works offline · synthetic data, nothing leaves the browser. Same numbers as the Snowflake app (health 32, ≈ 67 h to failure, Rs 16.8 L at stake, 36 of 43 failures caught).

YOURSTORY



1125



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THANK YOU

Nirantar · Team Aidhunik India (Paras Lehana, Abhinav Kaushik) · github.com/paras-lehana/nirantar
paras-lehana.github.io/nirantar/app (interactive demo, no login) (YouTube link in the submission form)